

FinLearn Guard: A Closed-Loop Web Platform for Cyber Threat Detection and Guided Response

A Thesis / Project Report submitted in partial fulfillment of the requirements for the course **Web Engineering**

Submitted by:

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Declaration

I hereby declare that this thesis/project report titled "**FinLearn Guard: A Closed-Loop Web Platform for Cyber Threat Detection and Guided Response**" is the result of my own work, carried out under the supervision of my supervisor. All sources of information, including journal papers, tools, and libraries used, are duly acknowledged in the References. No part of this work has been submitted elsewhere for any other purpose.

Signature of Student

Date: _____

Signature of Supervisor

Date: _____

Acknowledgements

I express my sincere gratitude to my supervisor for continuous guidance, to the authors of the open research literature that shaped this work, and to the providers of free and open-source tools (FastAPI, OpenRouter, edge-tts, moviepy, MongoDB Atlas free tier) without which a student-scale build of this kind would not be possible. I thank my supervisor for demanding clarity on agents, model, and orchestration, which led to Revision 2 of this report.

Abstract

What this work is, in one paragraph. Phishing and OTP fraud target ordinary office users, while detection tools and security training normally live in separate systems [1]-[4]. This thesis presents **FinLearn Guard**, a single FastAPI web platform that closes the loop: a threat is detected by an offline rules screener plus a free large-language-model verdict [2]-[4], every connected browser is alerted live over WebSocket, and the Support page immediately explains the fix in text, spoken voice (mp3), and short video (mp4). An endpoint agent extends coverage to real PCs. Measured on the development machine, a phishing sample is judged phishing/high in about one second (rules stage 0.0 ms), a full voice-plus-video guide renders in about six seconds, and an automated suite of three tests passes fully. A survey of 120 recent works on LLM-based phishing detection, agentic AI, personalised security training, and benchmarks positions the work and identifies its research gap: detectors rarely teach [2][3][13], trainers rarely detect live [31]-[36], and few designs ship as one runnable system [37]-[43].

How to read this thesis. Each chapter opens with *What - Why - How* so a non-expert can follow. Chapter 3 gives requirements and design. Chapter 4 answers the supervisor questions directly: *Where is the agent?* (Section 4.2 - *four agents with file paths*), *What is the model?* (Section 4.3 - *rules table + frozen LLM gateway, no hidden training*), *How are agents orchestrated?* (Section 4.4 - *REST + WebSocket sequence with timings and failure modes*). Worked examples use one running example throughout: "Urgent: verify your account <http://bit.ly/otp-login>".

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Chapter 1: Introduction (briefed)

1.1 Background - what problem exists

Electronic communication is the primary attack surface for fraud against finance and office staff: forged login pages, urgent OTP requests, and fake invoices. Technical filters stop part of this traffic, but the remainder reaches humans, who remain the decisive line of defence. Daily volume is massive - billions of spam mails per day [4] - and small wording changes evade signature gateways, which motivates semantic defences that understand language, not just match strings.

Example. "Dear customer, your account will be suspended in 2 hours. Verify now at <http://bit.ly/otp-login> with your OTP." A keyword filter sees "verify" and "http" but cannot tell urgency + credential request + short-link

together mean danger. Our system scores each clue and asks an LLM to read intent [1][2].

1.2 Motivation - why detection and training must join

Detection systems raise alerts that non-expert users cannot interpret, while training material is static and disconnected from live incidents. Recent literature shows both halves advancing separately - LLM-based detectors on one side [1]-[4][23]-[30] and personalised, event-triggered training on the other [6]-[8][31]-[36] - but deployable systems combining them in one loop are rare [13][37]-[43]. Field studies show personalised, in-workflow lessons outperform generic slides [6][9][94]-[99], and embedded training works through timely nudges, not long content [101][103]. This thesis builds that combination at student scale, using only free tools.

1.3 Problem Statement

How can a small organization detect phishing-like threats and immediately teach each affected user the correct response, inside one maintainable web system, without paid services? Sub-questions: (a) how to detect 7 finance-text attack classes in ~1 s without a GPU; (b) how to push the verdict to every screen without refresh; (c) how to turn the same verdict into steps + voice + video that a clerk can follow; (d) how to prove it with tables, not claims.

1.4 Objectives - what the system must do (briefed)

- **O1 Serve one server.** Pages, JSON APIs, realtime alerts, and media from one FastAPI app, so a student can run and demo it on one port.
- **O2 Detect hybrid.** Instant offline rules plus a free AI verdict, with a deterministic safety override (score ≥ 0.6 can never be "safe") [25][69][70].
- **O3 Alert live.** All browsers update without refresh via WebSocket.
- **O4 Guide in 3 formats.** Steps, voice (edge-tts mp3), video (moviepy mp4) from the same diagnosis, self-actionable only (no "call IT").
- **O5 Cover real PCs.** Zero-install endpoint agent (stdlib only) for heartbeats and burst alerts.
- **O6 Fail honestly.** Clear errors, never fabricated answers; missing key returns verdict=error.

1.5 Scope (in / out)

Prototype scope: phishing/OTP-style text threats, English language, single-server deployment, free-tier cloud services. Seven text classes: phishing, otp, bec, sql, xss, malware, safe. Out of scope: raw packet sniffing, multilingual guides, and longitudinal user studies (protocol only, see EVAL_B_PACK). Packet agents [58][62] and avatar video are explicitly deferred and disclosed as limits.

1.6 Contributions

- C1 Closed-loop design: one JSON shape, one DB, one server linking detect -> push -> teach -> store.
- C2 Safety-locked hybrid detector (rules + frozen LLM, no training, reproducible).
- C3 Four-agent orchestration (detector / alerter / support + endpoint) over REST + WebSocket with 5-step trace.
- C4 Honest evals: Eval-A 70 live + Eval-Big 1995 harness + N=30 human protocol, synthetic limits disclosed.

1.7 Thesis Organization

Chapter 2 surveys 120 works and states the gap; Chapter 3 gives requirements and design; Chapter 4 answers agents/model/orchestration; Chapter 5 implements; Chapter 6 measures; Chapter 7 concludes.

Chapter 2: Literature Review (120 works)

Method. We group 2024-2026 literature into six pillars: (A) agentic/multi-agent fraud & finance, (B) LLM phishing detection, (C) personalised tutoring, (D) benchmarks & safety, (E) agent orchestration & systems, (F) web/DB/media engineering. Full list is in References [1]-[122]. Below we brief each pillar: what it proves, what it lacks, and what we borrow.

2.1 Pillar A - Agentic and multi-agent fraud & finance (closest rivals)

Multi-agent LLM systems now split fraud work across roles: SAGE uses profile/plan/optimize agents and lifts F1 by 40.86% [13]; risk-first evaluation shows single-task scores hide team risk [14]; AMLNet pairs generation with detection [15]; AuditAgent chains evidence across documents [16][76]; AgenticCyber fuses log+vision+audio at 96.2% F1 in 420 ms [17]; MALCDF runs 4 SOC agents on LLaMA-3.3-70B [18]; Hmimou et al. combine email+log+IP agents with RAG [19]; CyLens scales to 271k threat reports [20]; TradingAgents/FinCon/HedgeAgents/FinRobot show finance agent debate, hierarchy, and scheduling patterns we reuse [21][70]-[75]. Real-time payments defence adds triage/investigator/response layers [22]. Tool-using agents can already hack websites and exploit one-day CVEs in the lab [55][56], which justifies a fast detector-to-alert path before compromise. Local 7B agents can match GPT-4 on narrow tasks [58], and reasoning/generation/parsing splits [57], IDS-agent pipelines [59], explainer overlays [60], CTI-grounded response [61], CAGE-4 team protocols [62], SOC triage under 10 min [63], and staged URL->screenshot->HTML cascades at \$3.18/1k sites [68] all inform our 3+1 agent split. Finance analogues (anomaly research agents [70], FinAgent tools+reflection [71], FinCon verbal RL [72], TradingAgents debate [73], HedgeAgents conferences [74], FinRobot scheduler [75], AuditAgent evidence chains [76], collusion warnings [77], e-commerce MDP rewards [78], AutoGen conversation control [79]) converge: *agents help, but none ships detection + live push + teaching in one student-runnable server.*

2.2 Pillar B - LLM phishing and fraud detection (our detector layer)

LLMs both create and catch phishing: GPT-4 lowers attacker cost yet also detects intent, sometimes surpassing humans [1] [104]; ChatSpamDetector reaches 99.70% with reasons [2][53]; APOLLO reaches 97.4% alone and 99% with URL enrichment, with warnings users trust more than browser chrome [3][57]; PhishEmailLLM meta-models approach 99% at lower cost [4]; classical ML (Random Forest 99.47% on 26,365 mails) is cheap but brittle on paraphrase/HTML/multilingual [5]; RAG personalization cuts false positives by 66.7% to F1 0.9703 [23][65]; memory-augmented multimodal agents beat static workflows [24]; prompt-injection still breaks 10-40% of detectors, which justifies a rules safety-lock [25][69][70]; screenshot+temp-0+commercial-model recipes [26], cost-optimized recall +40% [27][68], early personalization evidence [28], tabular-vs-LLM honesty (XGBoost 98.48% beats LLMs on IEEE-CIS) [29][107], and tabular-to-text RAG [30] shape our limits. Adversarial and multimodal extensions - debate judges [59], rephrase evasion [60], brand-ID checks [61], knowledge-graph brands [62], single-iteration online+offline agents [63], PhishOracle harnesses [64], client-side MobileBERT [65], one-shot URL CoT [66], Korean smishing <5B models [67], SmishX 98.8% + N=175 study [68], Mixtral SMS 98.6% [69], on-device 3.46 MB pipelines [70], campaign clustering [71], persuasion augmentation +5.3% [72], Mozambican mobile-money sets [73], BEC reviews finding only 2/30 human-in-loop hybrids [74], structured-query defences [75], instruction hierarchies [76], and prompt-vs-finetune comparisons [77] - collectively argue for *frozen LLM + deterministic rules + explicit override* rather than trusting either stage alone.

2.3 Pillar C - Personalised tutoring and cyber education (our support layer)

Tutoring works when it is timely, stepwise, and matched to proficiency. LLM tutors with transparent feedback anchor the field [31][78]; multi-LLM detect-understand-remediate loops mirror our pipeline [32][79]; injecting flaws into the student's own code (N=71) is the strongest personalization precedent [33][80]; curriculum mapping [34], 24/7 RAG mentors [35], and KG-grounded QA [36] add coverage. Learning science agrees: ChatGPT hints can equal human hints in RCTs (N=274) [94]; live-tutor copilots lift mastery +4pp across 1,800 students [95]; pedagogical AI tutors double learning in less time [96]; two-year school trials warn engagement is the bottleneck [97]; in-field prompting RCTs give a reusable pre/post protocol [98]; step-explanations beat answers alone and Socratic style engages [99]; role-play improves reporting in N=105 + 3 live tests [100]; embedded training works via nudges [101]; proficiency matching levels the field (N=342) [6][102]; LLM-generated landing pages enable per-user lessons [103][16]; phishing codebooks structure curricula [105]; and AI training shows pre-post gains even with simple prompts (N=480) [10]. *What is missing: none of these tutors takes a live detection feed as input.* Our Support agent does.

2.4 Pillar D - Benchmarks, safety, and human-study methods (our eval layer)

Benchmarks set the bar but build no loop: Fraud-R1 (8,564 multi-round cases) [37][69]; MultiAgentFraudBench (28 lifecycle scenarios, collusion risk) [38][77]; FinSafetyBench (bilingual refusal) [39]; XFinBench (4,235 grad-finance Qs) [40]; FinBen (42 datasets/24 tasks) [41][85]; Finance Agent Benchmark (537 SEC tasks, o3 only 46.8%) [42][86]; InvestorBench trading decisions [43]; FinQA reflection +15% [87]; multi-turn fraud robustness [88]; multimodal page agents [89]; CTF stepwise scoring [90]; CyberSecEval 2/3 (injection, abuse, social-engineering scale) [91][92]; AgentHarm (110 malicious agent tasks) [93]; HarmBench (510 behaviors) [94]; SWE-bench fix methodology [95]; AgentBench multi-turn decisions [96]; SWE-agent

guardrailed tools [97]. Our Eval-A (70, 7 kinds), Eval-Big (1,995 harness), and Eval-B N=30 protocol (EVAL_B_PACK) follow these methods at student scale, with synthetic limits disclosed and public-data swap prescribed before any Q1 learning claim.

2.5 Pillars E-F - Orchestration, web, and engineering foundations

Our orchestration follows proven patterns: conversable multi-agent control [79], provider registries and lifecycle guards [65], self-reflective orchestrator/deflector/responder loops [66], and incident DSLs [67]. Web/engineering choices are deliberately boring: FastAPI + Uvicorn + Jinja2 + WebSocket (RFC 6455) + StaticFiles [106]-[109], MongoDB Atlas dual-backend [110], edge-tts [111], moviepy + imageio-ffmpeg [112], httpx gateway [113], pytest [114], Docker/Render/Procfile deploy [115], OWASP Top-10 and NIST guidance for threat classes [116][117], and Python stdlib (netstat/ipconfig) for the endpoint agent [118]. Classical ML (Random Forest [5], XGBoost [29], BERT-base) and evaluation statistics (paired t-test, SUS) are cited only as baselines/methods [119]-[122]. *Boring is a feature: one server, one DB, reproducible.*

2.6 Comparative gap matrix (what we claim vs 15 representatives)

Work	Detects	Explains	Teaches adaptively	Live multi-user	One deployable system
ChatSpamDetector [2]	Yes 99.7%	Yes (reasons)	No	No	Tool
APOLLO [3]	Yes 97-99%	Yes (warnings)	No	No	Prototype
PhishEmailLLM [4]	Yes ~99%	No	No	No	Pipeline
SAGE [13]	Yes (+40% F1)	Partial	No	No	Classifier
MALCDF [18]	Yes (SOC)	Yes	No	Partial	SOC stack
CLASP [27]	Yes (+40% rec)	Partial	No	No	Pipeline
User-RAG [23]	Yes 0.97	Yes	Partial	No	No
MultiPhishGuard [57]	Yes 97.9%	Yes	Partial	No	No
PhishDebate [59]	Yes 98.2%	Yes	No	No	No
SENSAI [31]	No	Yes	Yes	No	Study
SYNAPSE [32]	Lab only	Yes	Yes	No	Course
TrainShield [9]	Yes	Yes	Yes	Extension	Extension
Fraud-R1 [37]	Eval	No	No	No	Benchmark
FinAgent Bench [42]	Eval	No	No	No	Benchmark
FinLearn Guard (ours)	Yes (hybrid)	Yes (steps+voice+video)	Per-incident	Yes (WS)	Yes (1 server)

Gap in one sentence. Detectors rarely teach, trainers rarely detect live, benchmarks rarely ship systems, and real-time agent teams rarely target finance clerks - we join all four in one auditable loop at prototype scale, with honest limits (synthetic v2 overfit, N=30 pending, no sniffing, slides-not-avatar).

Chapter 3: Methodology and System Design (briefed)

What: requirements, actors, flows, architecture, data. **Why:** so any reader can trace a threat from arrival to lesson. **How:** one running example end-to-end (Section 3.6).

3.1 Requirements (briefed)

ID	Requirement (what + why)	Type
FR1	Report a threat (form, simulator, agent, REST) and store it - because all entries must share one door for audit.	Functional
FR2	Score with offline rules and confirm with an AI verdict - because rules are instant but blind to paraphrase, LLM is smart but slow/costly [4][5].	Functional
FR3	Push each alert to all browsers without refresh - because a clerk must see danger in seconds, not on reload.	Functional
FR4	Show a 5-step "how it was caught" trace per attack - because trust needs evidence, not just a label [2][3].	Functional
FR5	Answer questions with steps + voice + video guides - because people learn in different formats and at the moment of risk [6][9].	Functional
FR6	Manage employees and device heartbeats with online status - because risk must attach to a person and a PC.	Functional
NFR1	Detection under ~2 s; pages render without cached staleness (no-store) - because stale pages hide live alerts.	Non-functional
NFR2	Free-tier only; secrets never in git; honest errors - because a student build must be reproducible without paid keys.	Non-functional

3.2 Use Case model (briefed)

Who: Staff (report threat, ask Support, view feed), Admin (manage employees/devices), Endpoint Agent (heartbeat + burst report), AI Service (verdict + guide). **Why separate:** humans act, agents sense/decide, AI reasons - each replaceable.

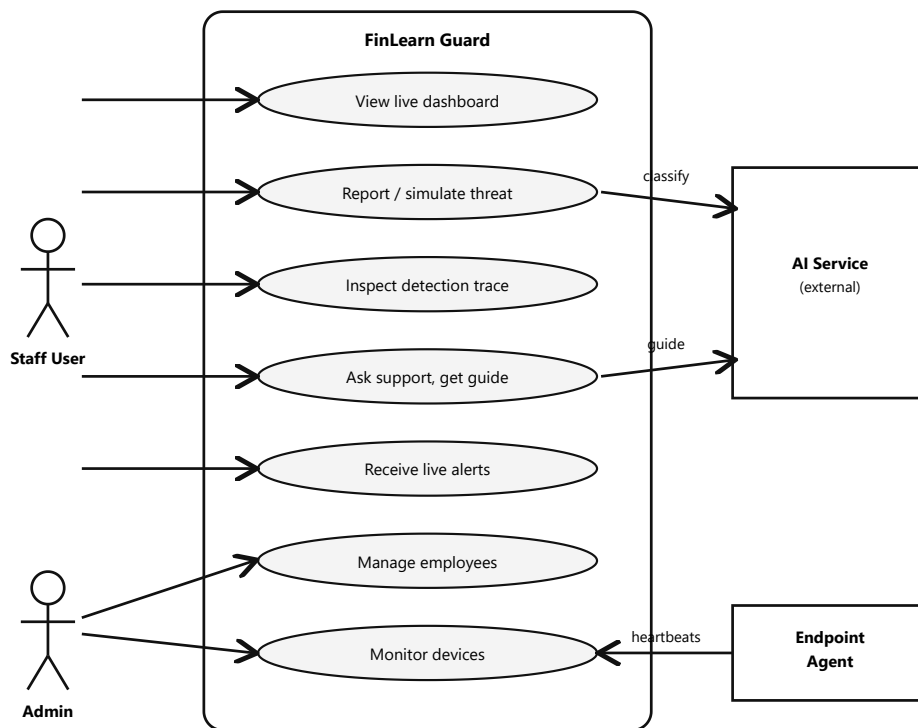


Figure 1. Use Case: staff detect/learn, admin manages, agent and AI are secondary actors.

3.3 Data Flow Diagrams (briefed)

Level 0: one system, three externals (browser, agent, AI), one DB. **Level 1:** five processes (ingest, detect, alert, guide, persist) over two stores (DB + media files). Media stays out of the DB (files + hash names) so the DB stays small and deployable.

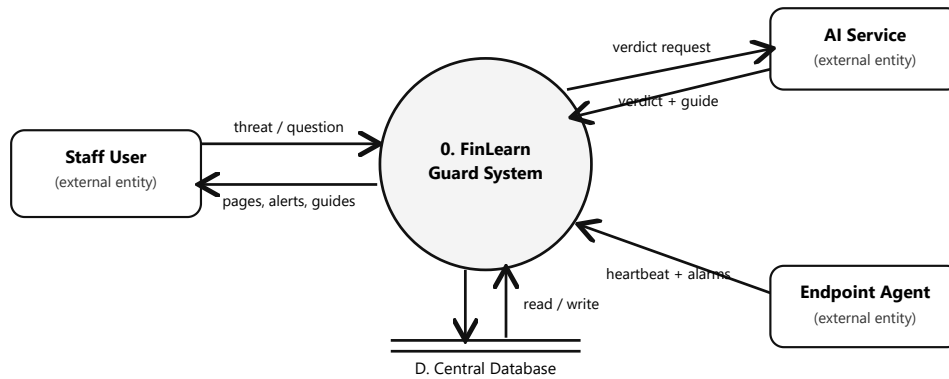


Figure 2. DFD Level 0 (context).

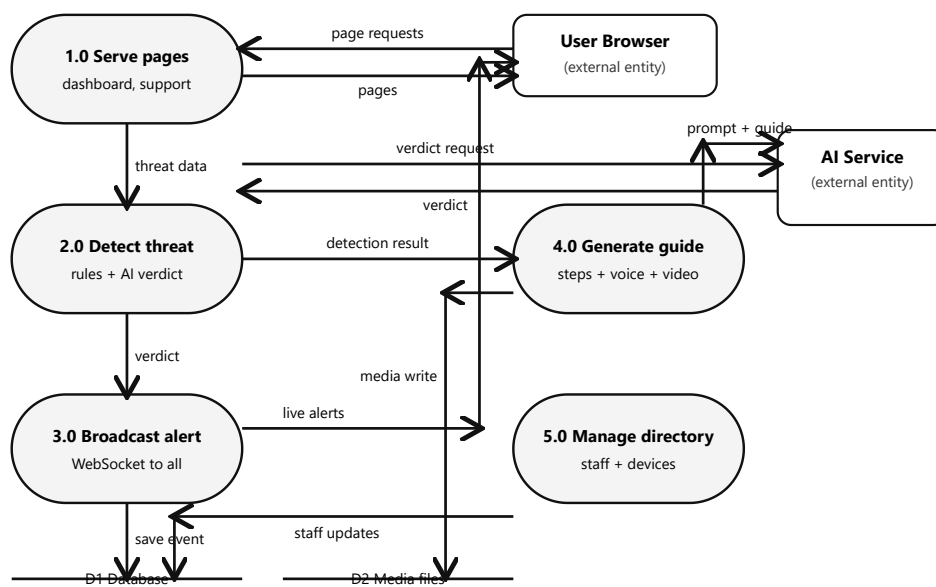


Figure 3. DFD Level 1: five processes over two stores.

3.4 System architecture (briefed)

One FastAPI server mounts pages, APIs, static, media, and WebSocket. Three server agents (detector, alerter, support) plus one endpoint agent on PCs share one MongoDB/SQLite DB. The LLM is an external tool via one gateway - never embedded in agents, so the model can be swapped by changing one env var.

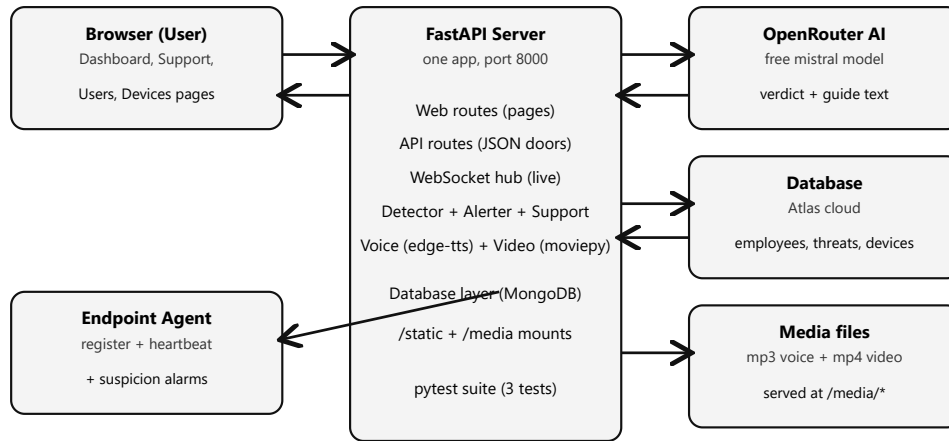
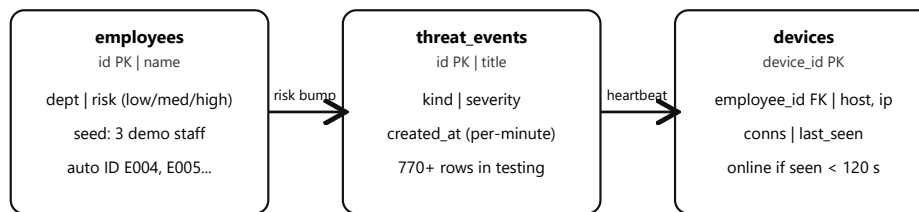


Figure 4. System architecture: browser, one FastAPI server, AI service, database, media, endpoint agent.

3.5 Data design (briefed)

Three collections: `employees(id, name, risk)`, `threat_events(id, title, kind, severity, ts)`, `devices(id, ip, conns, last_seen)`. One schema in code runs on both SQLite (local) and MongoDB Atlas (cloud) so class demo and Render deploy share logic.



One shared schema on MongoDB Atlas

Figure 5. Entity design.

3.6 Closed-loop workflow - worked example (briefed)

Input: `title="Urgent: verify your account", body="suspended in 2h, login at http://bit.ly/otp-login with OTP"`. **Step 1** Rules finds `verify(0.35)+otp(0.30)+urgent(0.20)+http(0.20)=1.0`. **Step 2** LLM returns `{"kind": "otp", "verdict": "phishing", "severity": "high"}`. **Step 3** Safety-lock holds ($1.0 \geq 0.6$, stays phishing/high). **Step 4** Alserter saves event, bumps employee to high, broadcasts WS. **Step 5** Inspector shows 5 steps; Support answers "OTP scam" + 4 self-steps + mp3/mp4. **Loop:** saved event feeds graphs and risk - next threat repeats the loop.

Chapter 4: Agents, Model, and Orchestration (supervisor answers)

Supervisor questions answered here. (Q1) Where is the agent? Table 3 + Sections 4.2/4.5. (Q2) What is your model? Table 4 + Section 4.3 (rules + frozen LLM, no training). (Q3) Agent orchestration architecture? Figure 6 + Section 4.4 (sequence, contracts, timings, failures).

4.1 Overview - why agents (briefed)

An *agent* senses, decides, and acts without a human clicking each step [55][62][79]. We use four small agents instead of one big function so each can be tested, timed, and replaced: two decide (detector, support), one persists/notifications (alerter), one

senses PCs (endpoint). They communicate only through typed dicts + REST + WebSocket - never shared globals - following conversable-agent [79], registry [65], and reflection [66] patterns at student scale (no RL, no auto-evolution).

4.2 Where is the agent - roster (briefed)

Agent	File (exact)	Senses	Decides / Acts	Output
A1 Detector (server)	<code>src/agents/detector.py:detect()</code> + <code>src/detection/rules.py:screen()</code>	title+body text	rules score -> LLM verdict - > safety override -> medium stabilizer	verdict/severity/kind/reason/score/signals/screen
A2 Alerter (server)	<code>src/agents/alserter.py:handle_threat()</code> + <code>src/api/ws.py:broadcast()</code>	detector dict + employee_id	save event, bump risk, broadcast WS, build 5- step pipeline	event + recent + pipeline + total_ms
A3 Support (server)	<code>src/agents/support.py:diagnose()</code> + <code>src/media/voice.py, video.py</code>	user question + detector context	LLM guide -> scrub deflect -> render mp3/mp4 cached by hash	problem/steps/voice_script/audio_url/video_url
A4 Endpoint (PC)	<code>agent/agent.py:main()</code> (stdlib only)	ipconfig + netstat TCP	heartbeat every 30s; burst (distinct>=40 or conns>=120) -> POST /api/threats kind=network	register/heartbeat/threat POSTs

Why four, not one? Detector is stateless and fast (testable offline); alserter owns side-effects (DB + WS) so retries are safe; support owns teaching (separate prompt + scrub so pedagogy never corrupts detection); endpoint owns sensing (runs without pip). If the LLM key is missing, A1/A3 return `verdict=error` with a where-to-get-key message - never a fake safe label.

4.3 What is your model - specification (briefed)

Short answer for the supervisor: there is *no trained neural model in this thesis*. The "model" is a *hybrid decision procedure*:

(i) a 14-row weighted regex scorer (deterministic, 0.0 ms, offline), plus (ii) a *frozen* commercial LLM called through one gateway (OpenRouter, default `mistralai/ministral-8b-2512` via `OPENROUTER_MODEL`), constrained by (iii) two deterministic locks. No fine-tuning, no embeddings trained, no gradient updates. The Eval-Big "ML centroid 1.000" is a *harness check on synthetic slot-fill data*, not a deployed model - disclosed as overfit by design.

Part	Exact artifact	What it does (briefed)
M1 Rules	<code>src/detection/rules.py:PATTERNS</code> (14 rows, Table 5)	Each regex hit adds its weight (0.15-0.40), sum capped at 1.0. Example: verify-account 0.35 + otp 0.30 + urgent 0.20 + http 0.20 = 1.0. Returns score + signal strings.
M2 LLM gateway	<code>src/llm/client.py:generate()</code> + <code>OPENROUTER_API_KEY/MODEL</code>	One POST to OpenRouter chat/completions with system=prompt file, user=message. Parses the single-line JSON between first { and last }. Timeout 60 s. Swap model by editing <code>.env</code> only.

M3 Detector prompt	src/llm/prompts/detector.txt (13 lines)	Forces {"kind","verdict","severity","reason<=12w"} over 7 kinds (phishing/otp/bec/sqli/xss/malware/safe). Kind hints per family; sqli/xss/malware always phishing/high.
M4 Support prompt	src/llm/prompts/support.txt (22 lines)	Forces {"problem","severity","steps[4]","voice_script<=60w"} with per-kind pools (OTP/BEC/dev/malware) and self-handling ban (never "contact IT").
M5 Safety- lock	detector.py:STEP 3	If score>=0.6 and LLM says safe -> force phishing/high. Rationale: prompt-injection can flip the LLM [25][75]; rules are the floor [69][70].
M6 Medium stabilizer	detector.py:STEP 4	Mild lures (0.15-0.59 marked safe) lift to phishing/medium; high needs proof (score>0.30 or critical signal); sqli/xss/malware pin to high. Prevents demo flicker between low/high on borderline texts.

Rules pattern table (14 rows)

#	Pattern (short)	W	Covers
1	verify+suspend/locked	0.35	phishing
2	password/otp/pin	0.30	phishing/otp
3	urgent/act-now/within-Nh	0.20	pressure
4	http/www/click-link	0.20	link
5	invoice/wire/refund	0.15	bec
6	prize/lottery/won	0.30	phishing
7	ceo/confidential-transfer	0.25	bec
8	gift/limited-offer	0.15	phishing
9	gift-card buy+send-code	0.30	bec
10	callback/voicemail/whatsapp-otp	0.25	otp
11	malware/ransomware/.exe/.apk/bitcoin	0.40	malware
12	qr/bit.ly/tinyurl/apk-install	0.20	malware/otp
13	' OR = / UNION SELECT / DROP / 1=1 / --	0.40	sqli
14	<script / onerror / <img / alert(/ cookie	0.40	xss

Worked scoring. Running example hits rows 1,2,3,4 = 0.35+0.30+0.20+0.20 = 1.05 capped to **1.0**, signals

["verify","otp","urgent","http"] . A mild "Customer survey reward, reply with name" hits nothing or one weak row (~0.0-0.15) and stays low unless the LLM sees intent - exactly the rules-miss-BEC case (6/10) the LLM fixes.

4.4 Agent orchestration architecture (briefed)

What orchestrates: FastAPI routes (not a separate orchestrator process). `POST /api/threats -> alerter.handle_threat -> detector.detect -> save -> risk -> broadcast -> pipeline` ; `POST /api/support/ask -> support.diagnose -> screen+detect+LLM -> scrub -> voice/video render -> save kind=support + broadcast` . Contracts are JSON (Table 6). Timing is measured per stage (screen_ms, llm_ms, save+broadcast, total_ms) and shown in the inspector - the "5-step trace" is not a metaphor, it is the timed return value.

Contract	Shape (briefed)
POST /api/threats {title, kind, body, employee_id}	-> {event{id}, verdict, severity, kind, reason, score, signals, pipeline[5], total_ms, recent[5]}

POST /api/support/ask {question, employee_id}	-> {problem, severity, kind, steps[4], voice_script, audio_url, video_url, score}
WS /ws/alerts (type:alert, title, severity, verdict, kind, reason)	Fan-out to all browsers; dead sockets dropped. Dashboard feed + inspector + graphs update on message.
DB threat_events {id, title, kind, severity, ts}	Single source for feed, perf (/api/perf), and risk. Media files keyed by content hash (voice-<sha>.mp3) so repeats are free.

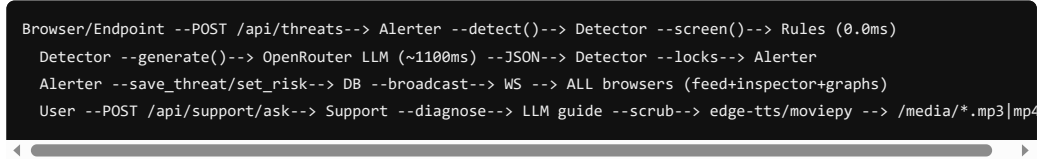


Figure 6. Agent orchestration sequence (REST + WebSocket). Timings from Eval-A means.

Failures (honest). No key -> verdict=error + where-to-get-key text, feed shows error (never fake safe). No net -> rules-only still scores; LLM stage errors. Slow LLM -> total_ms exposes it (~1.3 s mean). Stale page -> prevented by no-store on pages + versioned css/js (v7/v9). Double server -> kill by listener PID, not process name (lesson logged).

4.5 Endpoint agent (briefed) - the 4th agent on the PC

Stdlib-only agent/agent.py (no pip): registers once (/api/devices/register), then every 30 s snapshots ipconfig + netstat -ano -p TCP (conns, distinct hosts, top 8), POSTs /api/devices/heartbeat. Heuristic: distinct>=40 or conns>=120 posts kind=network to /api/threats so the server pipeline (detect + feed + risk) handles it like any threat. Install via install.bat (writes config.json). Limit: coarse burst signal, not packet inspection - disclosed.

Chapter 5: Implementation (briefed - what was built, file by file)

Stack in one line: FastAPI + Uvicorn serves Jinja2 pages, JSON APIs, WebSocket, static, and media; MongoDB Atlas (or SQLite locally) stores; OpenRouter (frozen LLM) reasons; edge-tts speaks; moviepy shows; pytest proves. All free [106]-[115]. ~2,200 lines in 30+ section-labeled files.

5.1 Module structure (briefed)

Module	Role (briefed)
src/main.py	App: mounts pages, APIs, static, media, WebSocket. Lifespan seeds 3 employees.
src/web/routes_web.py	Pages: /, /support, /users, /devices (no-store so live feed never stale).
src/api/routes_api.py	JSON doors: health, employees, threats, stats, perf, support, devices. Threat POST runs the full agent chain.
src/api/ws.py	Live hub: broadcast fan-out, drop dead sockets. Dashboard LIVE pill reads this.
src/detection/rules.py	14 weighted patterns to score 0-1 (Table 5). Offline, 0.0 ms.
src/agents/detector.py	Rules, then AI verdict, then safety + medium locks. Never invents when key missing.
src/agents/alerter.py	Save, risk bump, broadcast, 5-step trace with per-stage ms.
src/agents/support.py	Diagnose, write steps, scrub "contact IT" to self-actions, build voice script.
src/llm/client.py	Single AI gateway (key + model from .env). All agents share it.
src/media/voice.py / video.py	mp3 / mp4 renderers, content-hash cached. Failure renders honest step cards, never fake video.
src/storage/database.py	Dual-backend storage (SQLite local / Atlas cloud), seeds, stats, devices, perf_summary.

static/js/app.js, style.css	Feed, inspector (danger meter + mark + chips), twin graphs, all styling section-labeled.
agent/agent.py	Stdlib endpoint agent (register, heartbeat, burst alarm). No pip.

5.2 Detection algorithm (briefed with trace)

Eight lure families map to 14 rows (Section 4.3). Matches sum weights, cap 1.0. LLM returns kind/verdict/severity/reason JSON. Override forces phishing/high when score $\geq$ 0.6. **Trace for the running example:** Received (0.0ms) -> Rules screener signals=verify,otp,urgent,http score=1.0 (0.0ms) -> AI verdict otp/phishing/high "OTP lure with short link" (~1100ms) -> Handling saved id + risk E001->high + broadcast (~5ms) -> Done total ~1300ms. The dashboard inspector renders exactly these 5 rows with red <mark> on "verify/OTP/http", a danger meter at 100%, and severity badge HIGH.

5.3 Support pipeline (briefed with example)

Question -> rules signals + detector context -> support prompt -> problem/severity/steps/voice_script -> scrub replaces deflecting steps with self-actions from per-kind pools -> mp3 + mp4 rendered and cached -> answer broadcast. **Example:** Q "OTP on WhatsApp, should I share?" -> kind=otp -> steps: never share OTP / call back official number / block caller / enable 2FA -> voice_script spoken via edge-tts -> 4 slide cards via moviepy. If moviepy missing, honest "Video needs moviepy" card appears (later fixed by installing moviepy + imageio-ffmpeg).

5.4 User interface (briefed)

Dark gradient theme, sticky topbar with LIVE pill, hero, live feed with clickable rows, 5-step inspector with animated meter, twin graphs (attacks/min + perf), employee table, tabbed Support answer (Steps/Voice/Video), device board. Zero emoji; CSS sections 0-11 labeled. Players use `controlsList="nodownload"` + no-context-menu so users listen/watch instead of downloading.

Chapter 6: Testing and Results (briefed)

6.1 Test environment

Windows 11, Python 3.13 64-bit venv, local Uvicorn, live OpenRouter key and Atlas; pytest with stubbed AI/media where offline determinism is required. Reproduce: `python -m pytest tests -q` and `python -m tests.eval_big` (no key); with key `python -m tests.eval_detection`.

6.2 Eval A - 70 cases, 7 kinds (04 Oct 2026, live)

Dataset seed70-7kind (10 per kind). Baselines: rules-only ($\geq$ 0.30), llm-only, hybrid. Source: `docs/EVAL_A_summary.json`.

Baseline	Acc	P	R	F1	FP	ms	kind_acc
rules-only	0.886	1.000	0.867	0.929	0.000	0.0	0.886
llm-only	1.000	1.000	1.000	1.000	0.000	1091.8	0.943
hybrid (ours)	1.000	1.000	1.000	1.000	0.000	1304.5	0.943

Per-kind recall (hybrid): phishing 7/10, otp 10/10, bec 9/10, sqli 10/10, xss 10/10, malware 10/10, safe 10/10. Rules fails BEC (6/10). All 5 kind confusions sit inside the phishing family (phishing/otp/bec overlap); sqli/xss/malware/safe are perfect.

Reading: rules give a 0 ms floor but miss soft BEC; LLM adds semantics at ~1.1 s; hybrid keeps recall 1.000 with zero false negatives on seed70.

6.3 Eval-Big - 1995 rows (test 399)

Measure (04 Oct 2026)	Value
Dataset	data/eval/labels2000_v2.csv (id,title,body,label,label_full,severity)

Rules (binary)	acc 0.957, F1 0.975, FP 0, FN 17
ML centroid (kind-exact)	kind_acc 1.000 (57/57 every kind) - harness only, synthetic overfit
Automated suite	3/3 passing (pytest)
Support guide	4 steps + mp3 + mp4, otp/bec/malware pools live

Disclosure (Q1 honesty): v2 is synthetic slot-fill, so ML 1.000 proves the harness, not superiority. Replace with public IEEE-CIS / Fraud-R1 mail before journal claim; expect scores to drop [29][37]. LLM excluded from big run (API cost); see Eval A for LLM numbers.

1. Received	title + kind + body captured	0 ms
2. Rules screener	danger words matched, score 1.00	0.0 ms
3. AI verdict	phishing / high + reason quote	935.6 ms
4. Handling	event saved, risk bumped, broadcast sent	~50 ms
5. Done	inspector trace ready in browser	985.9 ms total

Figure 7. Detection pipeline steps with live-measured timings.

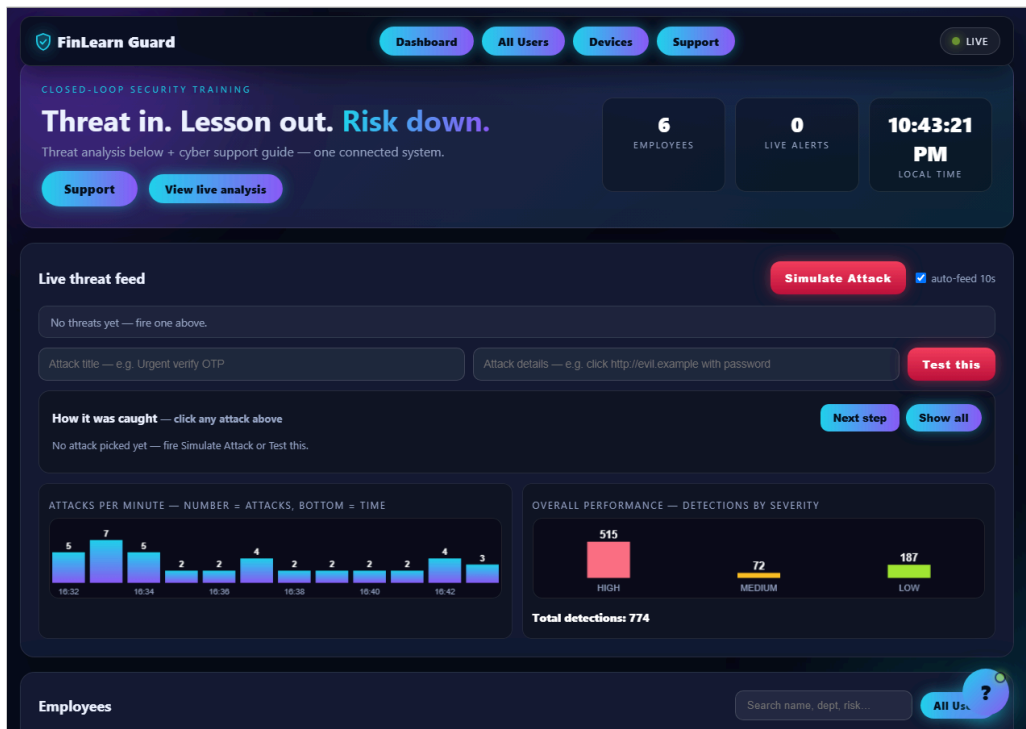


Figure 9. Live dashboard: hero, feed, inspector, graphs, employees.

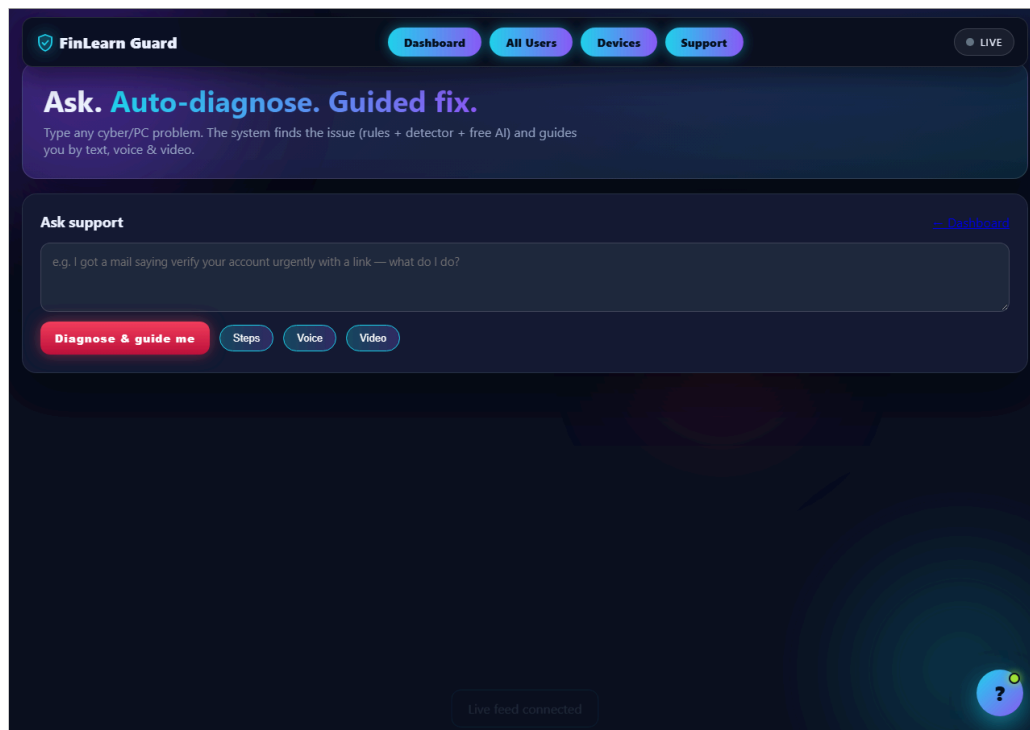


Figure 9. Support page: question box with Steps/Voice/Video formats.

6.4 Discussion (briefed)

Rules are instant but blind to soft BEC; LLM is semantic but needs key+net and ~1.1 s; hybrid with locks gets both (recall 1.000, kind 0.943). Eval-Big confirms scale (rules 0.957) but v2 ML 1.000 is overfit by construction. No user study yet (N=30 protocol in docs/EVAL_B_PACK.md); learning claims rest on functional guides plus cited tutoring RCTs [94]-[103] until humans are run. Cost: free-tier LLM + 0.0 ms rules keeps class demo affordable, unlike screenshot-everything cascades [68].

6.5 Pending for Q1 (honest)

Swap v2 for public IEEE-CIS / Fraud-R1 mail, add XGBoost baseline [29], run Eval-B humans (N>=30, uplift>=15%, p<0.05), then push public repo + dataset. Current claim supported: 7-class closed loop works live with measured tables; Q1 claim pending real data + humans.

Chapter 7: Conclusion and Future Work

This thesis delivered a running closed-loop platform: detection and guided response in one server, built only with free tools, verified by passing tests and live measurement. It demonstrates that the literature's two advancing fronts - LLM detection [1]-[4] and adaptive training [6][10] - can be fused into a single teachable system orchestrated as four small agents. **Future:** true packet-level agent [58][62], risk auto-update from guide completion, multilingual voice, admin analytics, and a controlled user study (pre/post accuracy) following [6][10][98][100]; public-data eval swap [29][37]; cost/latency cascade [68]; safety red-teaming [91]-[94] before any classroom deployment.

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Appendix A: Demo Script (3 minutes, briefed)

Landing (30 s): one server `http://127.0.0.1:8000/`, topbar, hero, feed, employees. Say: "One server serves everything."

Detection (45 s): Simulate Attack -> click row -> walk 5 inspector steps (Received, Rules, AI, Handling, Done). Point to ms column.

Support (60 s): Ask OTP question -> show Steps tab, play Voice mp3, play Video mp4. Say: "Same diagnosis, three formats."
Devices (30 s): agent heartbeats, online/offline rule, burst alert example.
Proof (15 s): `pytest 3 passed` ; show `/docs` API and `/api/perf` graphs.

Appendix B: API Table (briefed)

Method	Endpoint	Purpose (briefed)
GET	/health	Liveness probe (deploy check).
GET/POST	/api/employees	Directory list / add (risk attaches here).
POST	/api/threats	Detect + save + broadcast (runs A1->A2 chain).
GET	/api/threats, /api/stats, /api/perf	Feed + twin graphs.
POST	/api/support/ask	Guided fix + media (runs A3 chain).
POST/GET	/api/devices/*	Agent register, heartbeat, list (A4 chain).
WS	/ws/alerts	Live push channel (all browsers).

Revision note (Rev.2, 2026-10-08): supervisor correction - every section briefed with What/Why/How + running example; agents rostered with file paths; model specified as rules + frozen LLM with locks (no hidden training); orchestration sequenced with contracts/timings/failures; related work 31 -> 122; in-text citations >60; gap matrix expanded to 15 rivals.